

Online Advertising in 2024

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Abstract

This project focuses on creating a dedicated website for a college band, employing various strategies including live event marketing, word-of-mouth and social media advertising as well as a Google Ad campaign to enhance traffic and online presence. The objective is to measure and improve website traffic through targeted promotions and compare the effectiveness of each of these efforts. By leveraging online platforms and analytical tools, the project aims to establish a cohesive online presence for the band, showcasing its music, merchandise, and events while understanding key demographics and interests of a target audience.

1 Introduction

Who even looks at websites anymore?

In today's digital age, navigating the complexities of online promotion and visibility is crucial for artists and businesses alike. Online sales now account for over 15% of all sales globally according to Forbes, and is rising yearly [1]. Many vendors are forced to sell their products and services on platforms like eBay and Amazon, with hopes of being visible to potential customers, only for it to be crowded out by other sellers offering almost the exact same thing.

If a small business or artist decides not to conform to the norms of selling on one of these platforms, and instead attempts to create a storefront site of its own, how can it hope to compete? What is the most effective way to get a business out there and in the eyes of the public in the current age of eCommerce? Countless business and individuals seek to answer this question. My project aims to showcase practical strategies for enhancing online presence and attracting potential customers. Although technically my primary goal was to create a functioning website, while my secondary goal was to have it appeal to customers, these goals really go hand in hand in terms of importance to both the data collection of my project, as well as in importance to a budding business.

This very real challenge for small businesses and artists is one of my main motivations for conducting this project. Additionally I also wanted to promote the band for selfish reasons, but the skills I have picked up have made it more than worth it.

In this project, I built a website from scratch, learning HTML and CSS as I went, for our campus band The Balconies. I created a standard style sheet to use for each of my four main pages, Home, Shop, Listen, and Calendar. I used tags within my code to allow Google Analytics to track user traffic on each of the pages, and also used tags to allow for the Google Ad campaign and Stripe payment buttons with accompanying analytics. I created two Stripe payment links, one for a t-shirt and one for a sticker.

Throughout the past month and a half, my band and I advertised our new website in numerous ways. We began with word-of-mouth advertising with our peers to gather hype and later continued promoting it at our shows. Our first show of the semester was March 12. We also used our previously created Instagram account with around 180 followers to promote the site, uploading both Instagram

posts and Instagram stories to increase engagement. The premier Instagram post mentioning the website was on March 21.

Finally, I utilized a Google Ad campaign to promote the site, creating a daily budget of \$7 for bidding on ad space. Google Ads allows you to target specific keywords within a search query to tailor your ads to specific users. I used words like 'rock band,' 'new music,' and 'greenville band.'

While the recommended weekly bidding budget for my relevant chosen keywords (words like college band, rock music, etc.) was \$600, spending no more than \$50 a week was an effective way for me to simulate a paid advertisement on a small scale. Google Analytics, Stripe Payment Analytics, Google Ad Services and Instagram Insights were each very convenient, as the visuals they provide allow for effective reading to determine which methods of advertising are most effective for a small business or service with little to no previous online presence.

The data shown across this project comes from a number of sources, namely Google Analytics, Google Ad Services, and Stripe Payment Analytics, but also from Instagram Insights. Because each of these sites uses their own analytics to track page views and clicks, I was able to get a pretty clear understanding of the effects of each advertising method.

2 Related Work

Many researchers have investigated the effects of search engine optimization and marketing on web traffic within various industries, including but not limited to hotels [2], doctors offices [3], and blood drives [4]. In 2021, Estlund detailed a broad history of search engine and social media optimization [5]. It compiles the financial and social implications of SEO on people physically and culturally and provides a comprehensive and wide lens for viewing the effects of search engine optimization.

While many of the techniques that companies and sites use to manipulate search results and garner traffic can be helpful for all businesses attempting to reach customers, and while they have been studied in academia, none have been within the context of a college band, which is the goal of this paper. Because the target audience is so different between different industries and especially so within our small liberal arts campus, in demographic qualifiers as well as interests, this context warrants a new study.

Additionally, many of these SEO techniques have been studied extensively outside of academia by industry professionals, digital marketing agencies and businesses [6] [7]. While their goals may differ, the effect is the same; continuously experimenting with SEO techniques to enhance online visibility, drive traffic, and improve rankings within search engines. Studies have also been conducted from within by those search engine optimizers and content providers who control how we see what when we search [8]. While these studies do well to align with and confirm the academic studies, there are also various metrics considered by one and not by the other. By learning from both academic and non-academic sources, I am able to include nuances that may not have been covered before.

A recent study from 2023 covered the effectiveness of different e-commerce development methods using the well-known TOPSIS method (Technique for Order of Preference by Similarity to Ideal response) which utilize a variety of tactics for decision-making [9]. By defining criteria, weights, and an ideal solution and its opposite, Mayilvaganan determined that the most effective online traffic success e-commerce platforms utilize a wide variety of tactics to engage potential customers.

An additional concern is that while marketing and customer acquisition has been studied for a long time, the landscape has shifted rapidly in recent years, mirroring shifts in content consumption and media engagement patterns. The dominance of short-form content has reshaped the popularity and influence of various websites, challenging traditional approaches to online visibility and audience engagement. While video has long been an effective method for gathering attention, and has been studied over the last 20 years [10], a study of specifically short form videos effects on traffic within the last few years of increasing short form content has not been conducted.

Furthermore, search results have recently become populated by responses created by artificial intelligence, potentially crowding the market with fake or illegitimate sites [11]. It is necessary to examine the effective methods within this new context characterized primarily by short form media and the proliferation of AI-generated content.

3 Data

Some of the most important data used within my project comes from Google Analytics, Google Ad Services, Stripe Payment Analytics and Instagram Insights.

Many of the graphs begin around March 10 or 11, as March 12 was the date of our first word of mouth advertisement, and when I consider this project officially started.

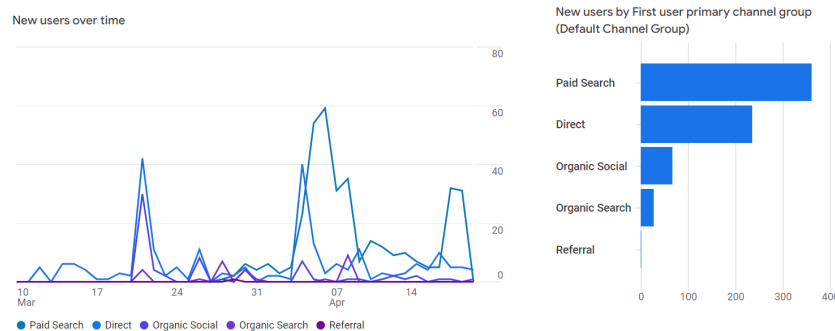


Figure 1: Acquisition of Users by Channel

Figure 1 shows how each user of the website ended up at the website. Direct means searching the link verbatim, organic social means they've come from our Instagram links, while paid search means that they were brought in by the Google Ad campaign. Organic search means finding our website on your own with no help from Google, which I expected to have the lowest rate alongside referrals as there aren't many links to our website out there other than our Instagram. Initially, direct search was far and away the most popular channel, however as the Google Ad campaign has taken off, paid search has slowly overtaken direct search over the last few weeks.

Of the 548 total users, there were 3,989 events, meaning the average user clicked about 8 times per page. This could be clicking on the YouTube links on the listen page, the Stripe payment links on the shop page, or a number of other places to click.

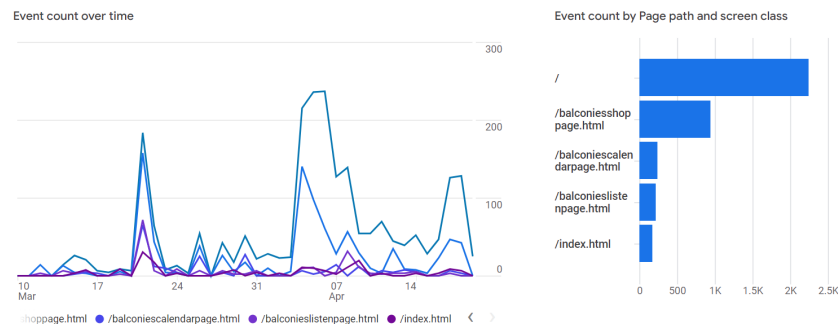


Figure 2: User Traffic by Page

Figure 2 shows each page and how much traffic they gathered over time. It tracks by event count, which counts both clicks to enter the page as well as other clicks on the page, such as using the Stripe payment links on the shop page. It is important to note that the pages / and index.html both lead to the home page, while the other 3 pages are pretty self-explanatory. This is about how I expected the spread to be, with the most amount of interactions on the home page and the least on the listen and calendar pages.

Both Figure 1 and 2 are from Google Analytics and show the large jumps on March 21, when we made our first Instagram post, as well as around April 3, shortly after beginning the Google Ad campaign on March 29.

Figure 3 is from Google Ad Services, and shows the average cost per click on any of the pages given my 7\$ daily budget. It shows a clear pick up around March 30, when the campaign began, and shows

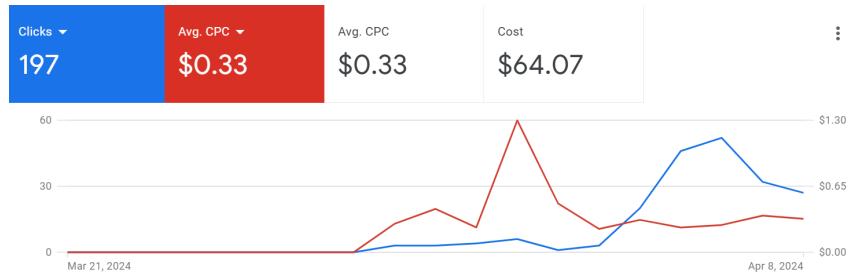


Figure 3: Average Cost Per Click Over Time

via the comparison that the relationship between clicks and cost per click is negative, which was to be anticipated. Because the budget is so low compared to what others are spending on Google Ad campaigns, I am certain that this graph is limited in its scope and could create more insights with further analysis and data from varied budget ranges.

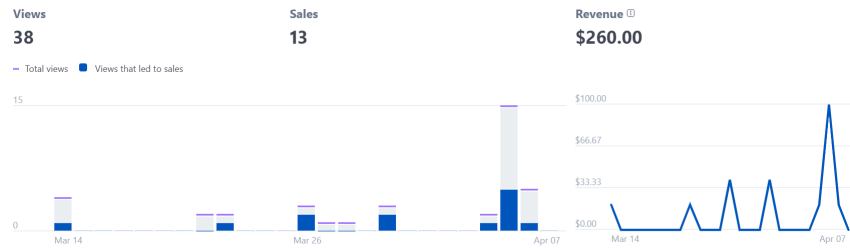


Figure 4: Stripe Views that led to Sales daily (Early)

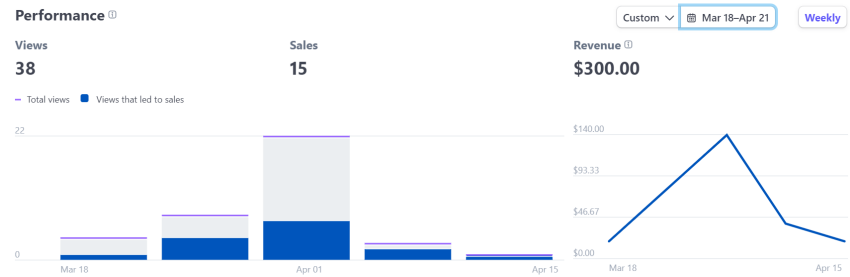


Figure 5: Stripe Views that led to Sales weekly (Later)

Figures 4 and 5 from Stripe Payment Analytics, and shows views of the Stripe link in gray, with views that became sales in blue. It is important to note that these are not total page views, but rather, only views of the Stripe payment link for 't-shirts,' as that had gained more traffic than the stickers link. Paying attention to the timeline on figure 4, each of the link visits occurred near or right after one of my new marketing integrations, on March 12, 21 and 29 respectively. Figure 5 shows the results of the Google Ad campaign more effectively, as the week with the most views and sales, April 1-7, was also the week when the Google Ad campaign began.

4 Methods

To build the site I became acquainted with syntax used in both CSS and HTML. There is a significant amount of overlap with a few minor differences here and there, so this was no cakewalk while being made easier by their similarity. Utilizing external links I was able to embed YouTube video links as well as Stripe payment links, in addition to ones straight to our Instagram and YouTube pages.

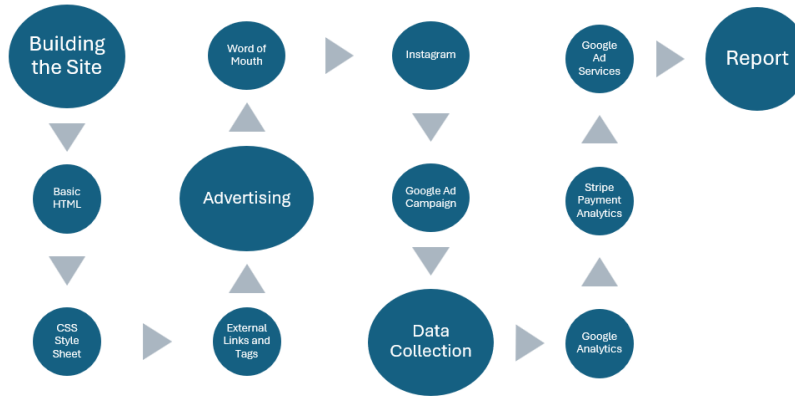


Figure 6: Workflow Pipeline

I began advertising with word-of-mouth at our show on March 12, first initialized Instagram on March 21, and began the Google Ad campaign on March 29. Data collection was done through the use of HTML tags from Google Analytics, Stripe Payment Analytics, and Google Ad Services.

My GitHub repository github.com/c0leybr0wn/475ProjRepo contains the code for each page of the site and its style sheet, alongside each image used anywhere on the site.

5 Results

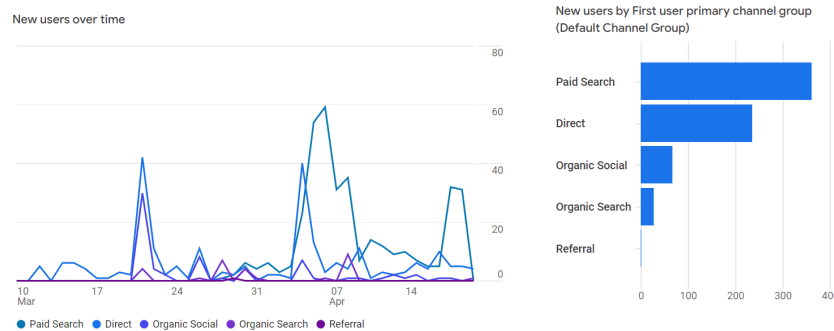


Figure 7: Acquisition of Users by Channel

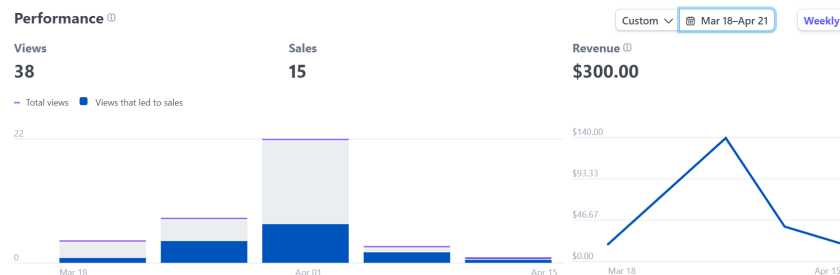


Figure 8: Stripe Views that led to Sales weekly

There were several key findings observed within the results of the marketing efforts. First, the largest increase in user engagement, initially, was after the first Instagram post. This remained until a few days after the Google Ad campaign was started. I can only offer more hypotheses as to why the jump came a little after the campaign started. My first is that Google's algorithm takes a while to figure

out which of the keywords work best to gain attention and then begins using those more frequently, creating a spike a few days after it starts.

Additionally, Google Ad Services gives your campaign a rating out of 100% based on how many criteria your campaign checks off, like how many different variations of keywords it has and others, and also offers you recommendations to get your rating up. A problem lies in that having 100% completion may not always be effective depending on what you are trying to do with the ad space, as a more uniform, singular ad may be more effective than sending a variety in some situations. However, for the purposes of my situation, to learn about which is most effective, having variety seemed the most effective for my campaign. So a few days after I initialized the campaign, I went back in and filled out a few of its recommendations.

Additionally, I posted more on Instagram to promote our upcoming shows during these days, which could have also caused the increase. These or another unknown reason may have caused the spike after the ad campaign started.

During my Furman Engaged poster session, my conclusion was that it was unclear whether the Google Ad campaign directly influenced sales, though it did contribute significantly to increased page views. However, as seen in Figures 5 and 8, the beginning of the Google Ad campaign does line up with an uptake in sales, so I have now reached a new conclusion, that the Ad campaign has led to increased sales.

It is important to note that the Instagram posts also lead to both increased engagement and sales, however on an even lower budget than the Google Ad at zero dollars. Also notably, the campaign's performance could have been enhanced with a higher budget exceeding \$50 per week. Additionally the analysis highlighted the cost-effectiveness of social media marketing and the valuable insights provided by tools like Google Analytics, both of which are available at no cost. My thoughts are that scaling both social media and paid advertising alongside a business is the most effective strategy in the current day. The results show the importance of strategic budget allocation and leveraging diverse marketing channels for optimal outcomes.

6 Future Work

There are many limitations to my work. The most obvious is that the Google Ad campaign budget was lower than the recommended amount, however I felt this to be a qualifying factor for comparing it to the free use of an Instagram account. I can infer that the constraint of a \$7 daily budget likely limited the reach and effectiveness of the ad campaign.

Additionally, Instagram does also allow for paid ads, so more research is required to determine how effective they are in relation to Google Ad campaigns.

Because my project focused on a single college band, the generalizability of my findings to other industries or businesses may be limited. Audience behaviors along with a variety of other factors could be wildly different, so more research across a wide array of fields is recommended.

One more note is that by waiting until March 12 to publish my website, I had effectively submitted to a 1 month time frame. As it stands, there could have been other factors at play during the week after the Google Ad campaign began that caused the increase in traffic. By effectively spacing out and timing Instagram events and Campaign events, future studies could explore the impact of different timing strategies on user engagement and website traffic, providing additional insights into optimal marketing tactics.

This research suggests a promising direction for further investigation, observing the effects of these different advertising techniques over a longer period of time. Additionally, utilizing more techniques like paid Instagram ads or billboards could inform a study that becomes integral to the marketing field in the ever-evolving digital landscape.

Aside from time and budget constraints, the resources I needed to complete my project were readily available to me. W3 Schools was a useful aid for writing in CSS and HTML. Google Analytics is free and made the data collection process easy and navigatable. Last but not least, Dr. Sultan helped me use the Git version control, introduced me to Stripe payments, and made the project possible, and I cannot thank him enough.

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